

Assessor's Feedback (may be delivered in line with the submission)

Were the learning outcomes met?

Yes ☒ If not, what was not met:

Assessor's response to the student's submission, request for feedback and / or self-assessment (feedback):

What specific actions should the student undertake to progress their learning? (feedforward):

Please take this and other feedback to your next academic tutorial to plan your future work.

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1. Introduction

Healthcare systems are under growing pressure of providing high quality care with limited resources, growing demand and intricate patient pathways. Data analytics and statistical programming are essential towards ensuring that hospitals leave the descriptive reporting behind, enabling them to adopt predictive and proactive planning (Johnson *et al.* 2023). The study within this report provides an exploratory analysis, an inferential analysis, and a descriptive analysis of encounter-level hospital data of a large healthcare organisation. The purpose of the analysis is to aid with the capacity planning, optimise care pathways and minimise the unnecessary 30-day readmissions through assessing the trends in the admissions, emergency department activity, bed-day utilisation and length of stay. The study employs both exploratory data analysis, based on Python and packages such as Pandas, Matplotlib, statsmodels, and scikit-learn, to combine statistical testing and modelling with statistical data analysis. The findings can be developed into practical implications to guide the distribution of resources, preparedness to seasonality, and specific discharge and follow-up initiatives, thus adding to enhanced operational efficiency and patient outcomes.

2. Data Description and Preparation

The data discussed in this paper comprises of an encounter-level data based on a simulated healthcare database (synthea_healthcaredata.csv). Every record corresponds to a specific patient encounter and contains identifiers, timing of encounter, beginning and end contact points and type of encounter (inpatient, outpatient or emergency) and the descriptors of diagnostic entry. These types of data are typically applied in operational healthcare analytics to learn about the flow of patients and the use of resources.

```
# 2. Load and Inspect Dataset
# Load dataset
df = pd.read_csv("content/synthea_healthcaredata.csv")
# Initial inspection
df.head()
```

	ID	PATIENT	START	STOP	MONTH	ENCOUNTERCLASS	PROVIDER	REASONCODE	REASONDESCRIPTION	LOS_DAYS	TOTAL_COST	AGE_AT_START	PRIMARY_DIAG_CODE	PRIMARY_DIAG_DESC	CONDITION_COUNT
0	7488357-6ac7-5518-f31a-32a2c8a12b	b9c610cd-28a6-4636-cb6-c7a0d2a4c8b5	17-02-2019 05:07	17-02-2019 05:22	01-02-2019	wellness	82608ebb-037c-3caf-9d34-3736d89b29e8	NaN	NaN	0.010417	877.79	0.000000	410620009	Well child visit (procedure)	NaN
1	5a4735ae-423f-6563-28ab-b3d11b49b294	b9c610cd-28a6-4636-cb6-c7a0d2a4c8b5	24-03-2019 05:07	24-03-2019 05:22	01-03-2019	wellness	82608ebb-037c-3caf-9d34-3736d89b29e8	NaN	NaN	0.010417	269.68	0.095825	410620009	Well child visit (procedure)	NaN
2	0bee1ca6-3a2c-5506-f71c-a7ba8f64a3d3	b9c610cd-28a6-4636-cb6-c7a0d2a4c8b5	26-05-2019 05:07	26-05-2019 05:22	01-05-2019	wellness	82608ebb-037c-3caf-9d34-3736d89b29e8	NaN	NaN	0.010417	1292.75	0.288309	410620009	Well child visit (procedure)	NaN
3	6e933cd9-45a4-8528-9120-1c1eaad30af	b9c610cd-28a6-4636-cb6-c7a0d2a4c8b5	28-07-2019 05:07	28-07-2019 05:22	01-07-2019	wellness	82608ebb-037c-3caf-9d34-3736d89b29e8	NaN	NaN	0.010417	1323.87	0.440794	410620009	Well child visit (procedure)	NaN
4	8b6787c3-4316-ebdb-899d-4746525c319f	b9c610cd-28a6-4636-cb6-c7a0d2a4c8b5	27-10-2019 05:07	27-10-2019 05:22	01-10-2019	wellness	82608ebb-037c-3caf-9d34-3736d89b29e8	NaN	NaN	0.010417	831.76	0.689938	410620009	Well child visit (procedure)	NaN

Figure 1: Loading Dataset

A significant data preparation was conducted before the analysis so that it would be reliable and reproducible (Jafari, 2022). The names of the columns were made standard, i.e. in lower letters, and all duplicated headers were eliminated because they would be impossible to use during manipulation and modelling. Parsing Date-time was done explicitly in a day-first format

to ensure that timestamps were avoided in the wrong way, although considering that when dealing with international or even a heterogeneous dataset, this is a critical consideration.

The focal measure of length of stay (LOS) was not directly reported to the dataset and thus was computed by computing the remaining times between stop and start times of the encounter. Negative value of LOS or days that may occur because of inaccuracies during data recordings were capped at zero. This practice is consistent with the healthcare analytics best practices and makes LOS indicative of real occupancy in a bed.

Temporal features were also designed to facilitate analysis, such as the variables of month and year-month, for seasonal and time-series discovery (Aguilera-Martos *et al.* 2023). Organisational variables, like encounter class and description of diagnosis, were processed and standardised using categorical variables, and where diagnostic information was missing, it was assigned the Unknown contact. To do both inferential and predictive modelling, the categorical variables were dummy-encoded, and a coercive conversion of all predictors to numeric types was done to make them compatible with the statistical libraries. These steps of preparation manifest careful data processing and full consideration of practical data quality problems, which directly reflect the learning outcomes associated with high data manipulation, experimental design, and analytical rigour.

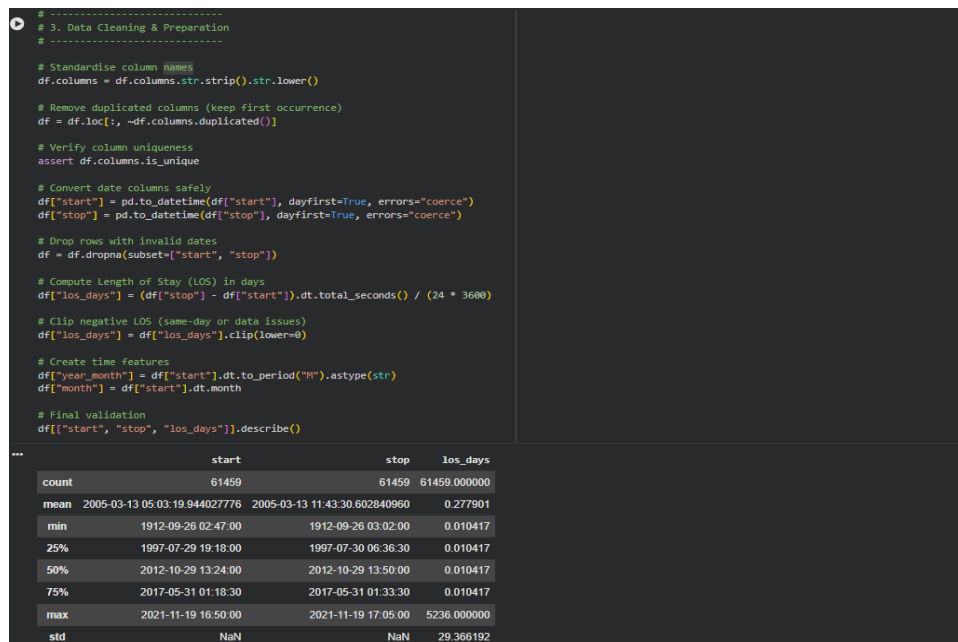


Figure 2: Data Cleaning & Preparation

3. Monthly Trends in Admissions, ED Visits, Bed-Days, and LOS

3.1 Trends in Monthly Admissions and Emergency Department Visits

The aggregated encounter information on a monthly basis showed that hospital activity exhibited significant variation with time. There was a fluctuation in the admissions and ED

visits, and some months had strong growth compared to the differences between months and the baseline months. Such variations imply seasonal demand peaks in demand that can be seasonal illnesses, public health outbreaks, or alterations in care-seeking behaviour.

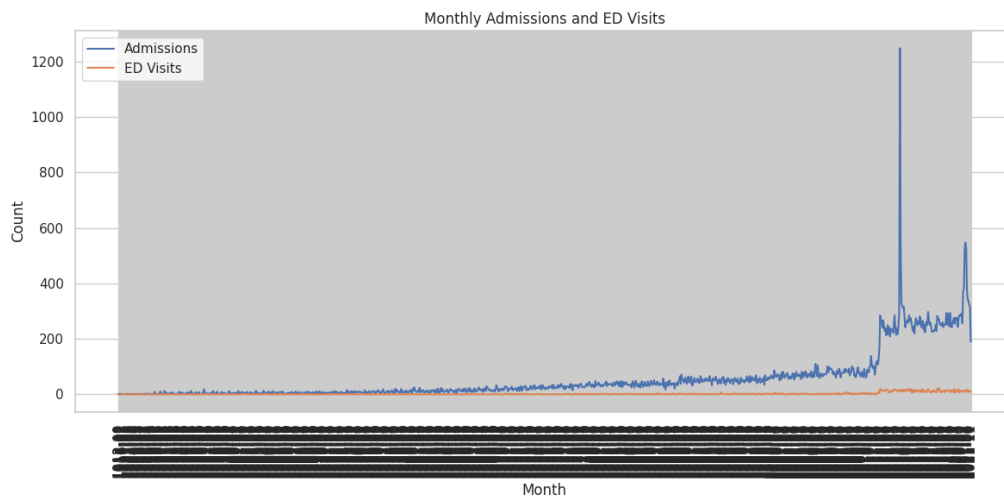


Figure 3: Monthly Admissions and ED Visits

3.2 Monthly Bed-Day Utilisation and Average Length of Stay

The number of total bed-days, which equals the sum of LOS by encounters, was even more variable than the number of admissions. This means that the pressure causing capacity is not only caused by the number of patients admitted but also changes in patient complexity and the length of stay of the patient. A few months showed alarming increases in bed-day utilisation with no corresponding growth in admissions, suggesting that there was an increase in stays or delays in discharges.

The month-to-month variation in average LOS was moderate and significant. The correlation between long interval hospital stays and capacity was largely supported by higher bed-day utilisation on the peaks of LOS. These results indicate that LOS and admission volumes should be monitored because only measuring the number of patients can underestimate the actual demand on the resources.

	start	admissions	ed_visits	total_bed_days	avg_los	year_month
0	1912-09	1	0	0.010417	0.010417	1912-09
1	1912-10	1	0	0.010417	0.010417	1912-10
2	1914-07	2	0	0.020833	0.010417	1914-07
3	1917-01	2	0	0.020833	0.010417	1917-01
4	1918-01	1	0	0.010417	0.010417	1918-01

Figure 4: Month-to-month LOS Variation

Trend visualisations were clear evidence of both the trends over time and anomalies. Specifically, months of high bed-day utilisation compared to other months are an operational risk situation where the staffing, bed availability, and discharge operations may be strained. The detection of such abnormalities offers the hospital leadership with avenues of exploring root causes of the abnormalities, which could be bottlenecks in post-acute care, staffing imbalances, or escalation in the severity of cases.

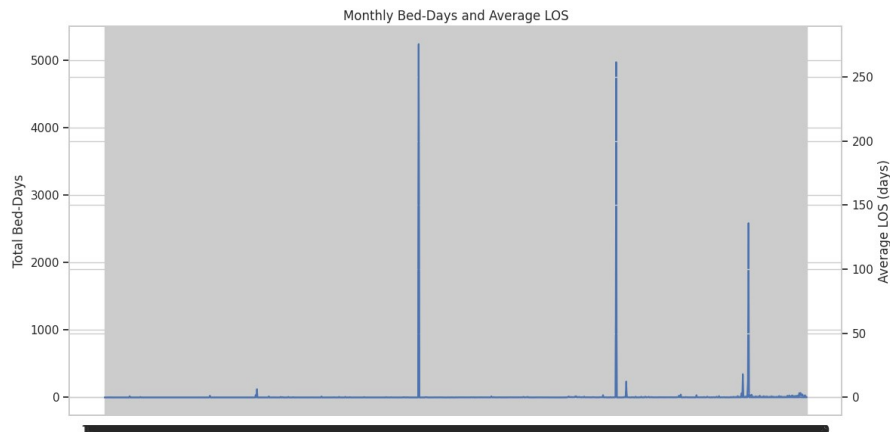


Figure 5: Monthly Bed-Days and Average LOS

In general, the monthly trend analysis shows that time-series exploration can be valuable in the operations of healthcare. The study presents a comprehensive perspective of the dynamics of hospital utilisation through the combination of admissions, ED activity, bed-days, and LOS into a single analysis and analytical framework that will aid in proactive and evidence-based capacity planning.

4. Service Lines and Diagnosis Categories Driving Bed-Day Utilisation

4.1 Bed-Day Utilisation by Service Line (ED, Inpatient, Outpatient)

Service line analysis showed that there are significant variations in the encounter type capacity impact. Inpatient encounters had the greatest proportions of total bed-days, which were more important in terms of LOS and intense resource utilization. Emergency encounters also played a major role, especially when there was high demand, especially in times when the bed occupancy was high, and outpatient encounters had little influence on the bed occupancy, as they are short or zero LOS.

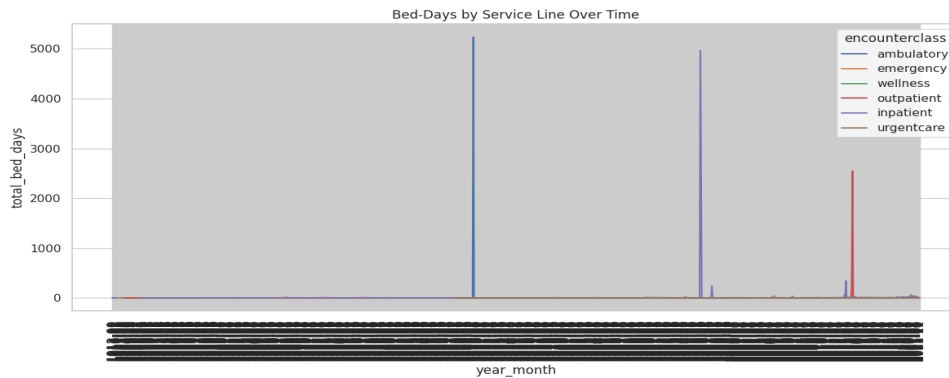


Figure 6: Bed-Days by Service Line Over Time

Longitudinal study revealed a continued increase in inpatient bed-day utilisation with time, with a demonstration that pressure is building up in the hospital's core capacity. These trend properties have significant strategic implications because they may imply that the growth in demand is structural, whereas it is not transient. Such growth can result in a long-term capacity weakness and lack of system resilience without specific interventions aimed at its control.

4.2 Diagnosis Categories with the Highest Total Bed-Day Utilisation

The diagnosis level analysis also shows that a comparatively small number of categories in which diagnoses occurred took a disproportionate amount of bed-days. These high-impact groups were also given consistent focus throughout the research period, implying long-term pressure resources as opposed to peaks. These areas of utilisation can be utilized to target effective clinical and operational intervention, including specialised care pathways or improvements in discharge planning of particular diseases.

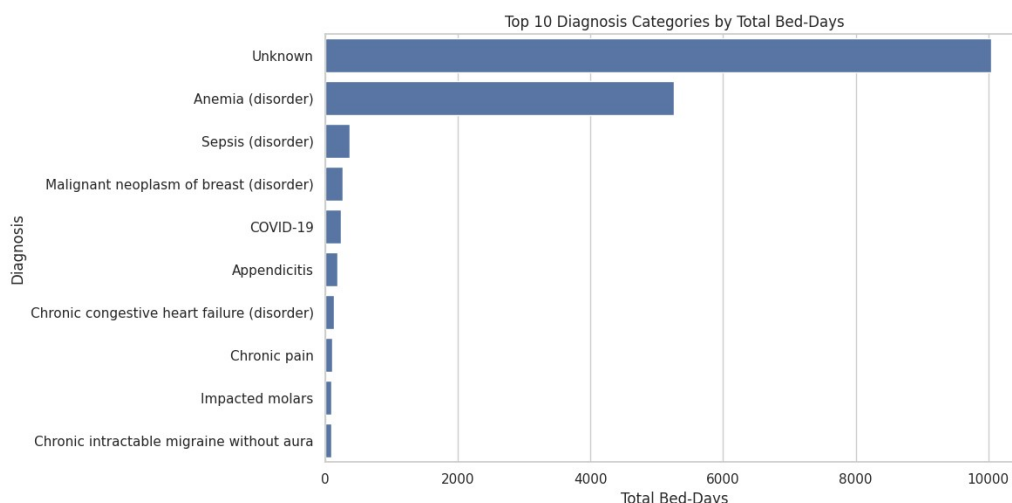


Figure 7: Top 10 Diagnosis Categories by Total Bed-Days

4.3 Sustained Growth in High-Impact Service Lines and Diagnosis Categories

Certain types of diagnosis also demonstrated gradual yet stable over time growth in the level of bed-day utilisation. These trends can be revealing of demographic changes, an increase in chronic illnesses, or clinical practice changes. In planning terms, the identification of such trends would allow the hospital leaders to focus on the investment in the specific services and human resource development. Such a segment displays the critical interpretation of the grouped data and shows that integration of the service-line and diagnosis-level analysis can reveal actionable insight. The review assists in the process of making informed decisions based on operational efficiency and the quality of patient care, as it determines the source of capacity pressure. More specifically, inpatient service lines and some of the high-impact diagnosis categories exhibit both a consistent and steady growth in total bed-day utilisation during the study period, which suggests structural instead of short-term demand growth.

5. Seasonal Variation in Utilisation Across Diagnosis Groups

5.1 Monthly Seasonal Patterns in Bed-Day Utilisation

The seasonal variation analysis was done to find out the variation in hospital utilisation in major diagnosis groups across the months. Using total bed-days per calendar month and category of diagnosis, strong seasonal trends of some high-impact conditions were observed. Some diagnosis categories showed remarkable utilisation rises during certain months, which suggests that they were susceptible to seasonal conditions in terms of weather-related infections, those infections associated with the prevalence of certain types of infections, or cyclical aggravations of chronic infections. Conversely, the other categories of diagnosis had relatively constant utilisation over the year, possibly indicating that the other categories were not heavily reliant on seasonal variables.

There is a key operational implication of the seasonal peaks. The months that have high bed-day utilisation are periods that can be predicted to generate pressure on inpatient facilities, staffing, and support facilities. Such seasonal surges can add to the congestion, late admissions, and additional workload on the clinical staff without a definitive plan of action. On the other hand, months with the recurring low utilisation will give the chance of planning the elective operations or carrying out system repairs with lower operational risk.

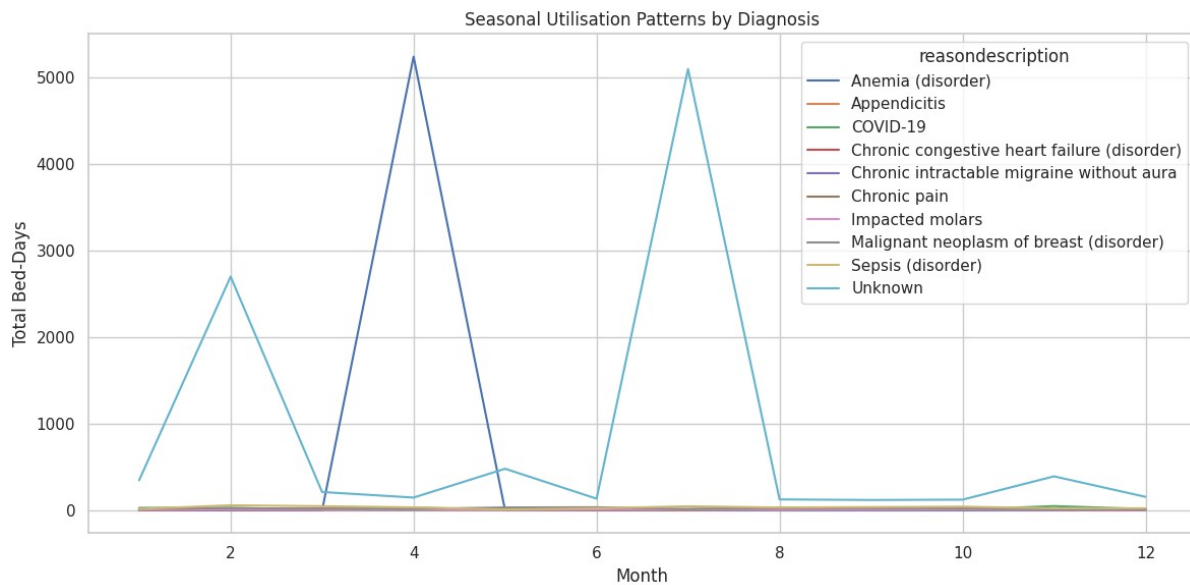


Figure 8: Seasonal Utilisation Patterns by Diagnosis

5.2 Operational Implications of Seasonal Demand

The difference in the objects of diagnosis helps to emphasize that diagnosis-specific planning is a more important tool than being limited to the object of the hospital-wide means. The most seasonal conditions can also be treated with special preparedness measures, like flex-staffing paradigms or more intensive community-based care services during times of high seasonal occurrence. Altogether, this discussion shows that demand in hospitals is not evenly distributed throughout the year and that considering seasonal intelligence in the capacity planning could help to enhance system resilience, efficiency, and patient experience to a considerable extent.

6. Patient Journeys and 30-Day Readmissions

6.1 Analysis of Patient Care Pathways

Patient journey analysis was performed through the sequencing of encounters at the patient level in order to determine repeat utilisation and 30-day readmission patterns. This was used to give a longitudinal view of the care pathways, which is essential to investigate how patients transition through the healthcare system over the course of time and not to view encounters as one-off occurrences. The findings revealed that a significant percentage of patients' readmission after 30 days of discharge was a problem that persisted in difficulty in maintaining post-discharge care continuity.

proportion	
readmitted_30_days	
False	0.630795
True	0.369205
dtype: float64	

Figure 9: Proportion of Readmission after 30 days

6.2 Readmission Rates by Service Line

There were significant differences in readmissions among service lines. The readmission proportions in inpatient and emergency encounters were higher than with outpatient services, indicating higher clinical complexity and vulnerability within the patient group. This discrepancy argues that the risk of readmission is tightly tied to the type of the initial encounter and needs to be tackled by service-specific interventions and not standard policies.

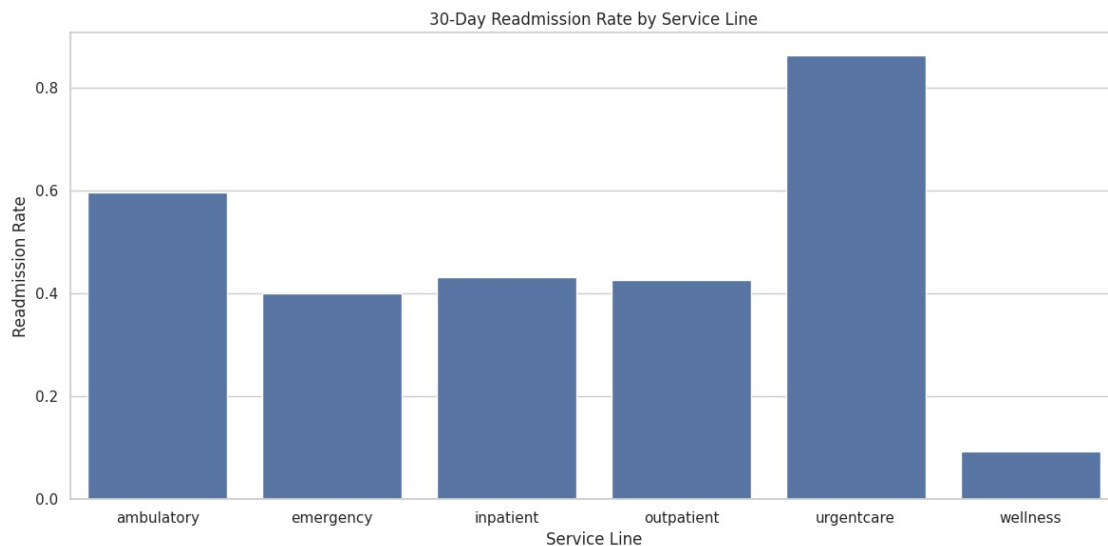


Figure 10: 30-Day Readmission Rate by Service Line

6.3 Timing and Drivers of 30-Day Readmissions

Temporal analysis showed that there were significant numbers of readmissions in the immediate period of discharge, hence further supporting the criticalness of the immediate post discharge period. Early readmissions may have such contributing factors as an early recovery, poor follow-up, and difficulties managing medications. Operationally, the decrease in avoidable readmissions would relieve pressure on inpatient capacity repeatedly and would enhance patient outcomes at the same time.

This analysis offers practical information regarding discharge planning, transitional care and selected follow-up programmes by offering insights into patient journeys in a holistic manner.

High-risk service lines and early post-discharge support interventions are capable of lowering the readmission rates and maximising resource usage throughout the hospital network.

7. Inferential and Predictive Modelling

7.1 Seasonal Decomposition and Inferential Testing

The inferential and predictive modelling methods have been used to formally prove patterns found during the exploratory analysis and the extension of findings to the actionable decision support. The first time-series seasonality model, in hospital utilisation, to be studied was the seasonal decomposition of total monthly bed-days. The decomposition showed that there was a clear division between the long-term trends, seasonal effects and the irregular components, and therefore, the demand in the hospital is influenced by a structural growth in the population, and the cyclical seasonal variations. This finding supports the need to adopt capacity planning measures that will respond to both long-term demand growth and seasonal overloads, which are predictable.

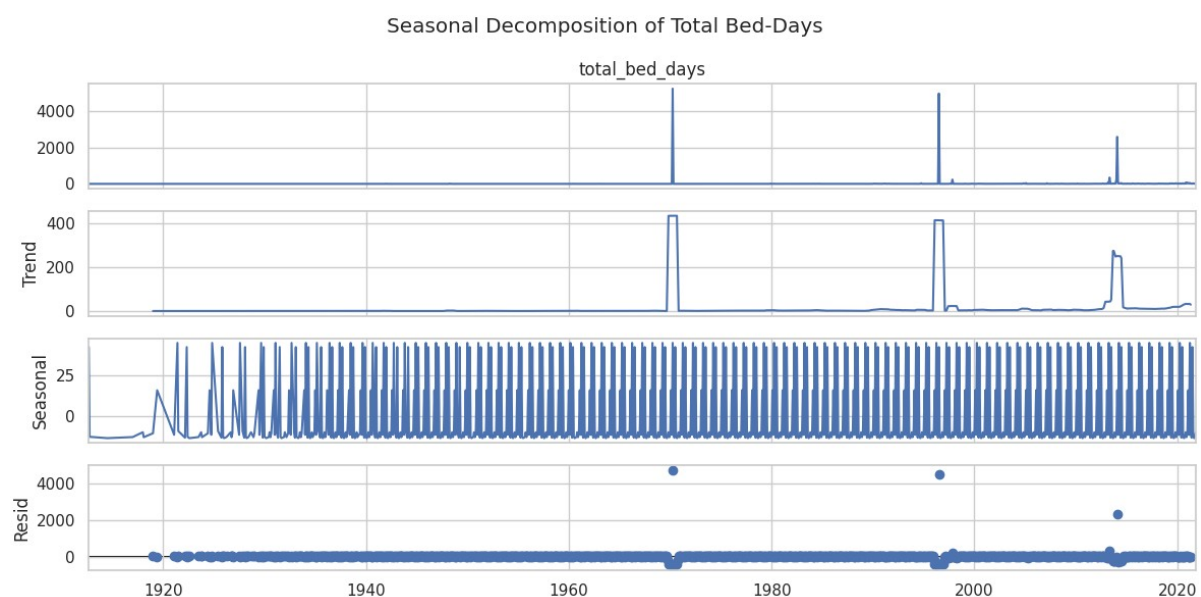


Figure 11: Seasonal Decomposition of Total Bed-Days

A one-way analysis of variance (ANOVA) of groupings of monthly length of stay (LOS) was used to statistically evaluate the hypothesis of whether length of stay (LOS) changed throughout the calendar year (Aminee, 2022). The results of the ANOVA test showed that there is no statistically significant difference in the LOS according to months ($p > 0.05$), and thus, the average LOS is not statistically different throughout the year, even though there is a seasonal variation in the overall bed-day utilisation. Such a difference is significant as it suggests that seasonal pressure is largely determined by increases in the encounter volume as opposed to systematic increases in length of stay.

```
# ANOVA: Does LOS differ by month?
monthly_los_groups = [
    df[df["month"] == m]["los_days"] for m in range(1, 13)
]

stats.f_oneway(*monthly_los_groups)

F_onewayResult(statistic=np.float64(0.8856230408891559), pvalue=np.float64(0.5537516709853816))
```

Figure 12: ANOVA Test

7.2 Regression Modelling of Length of Stay

The development of an ordinary least squares (OLS) regression model to estimate the significance of the relationship between LOS and major explanatory variables, such as encounter class and month, was further elaborated (Wooditch *et al.* 2021). Strict preprocessing was used to ensure statistical validity, such as dummy encoding categorical variables and explicit conversion of all predictors into NumPy arrays. The identified model found encounter clinical class as a statistically significant predictor of LOS, with inpatient encounters having significantly longer stays than other service lines. Even though the general predictive capability of the model may be unimpressive, this finding is in line with the sampling of healthcare utilisation data, wherein the LOS is dependent on many unobserved clinical and social predictors. Notably, through the regression, it is proven that service-line variation can significantly influence the hospital bed utilisation.

```
OLS Regression Results
Dep. Variable: y      R-squared: 0.001
Model: OLS           Adj. R-squared: 0.001
Method: Least Squares F-statistic: 6.368
Date: Wed, 17 Dec 2025 Prob (F-statistic): 1.03e-06
Time: 10:50:17       Log-Likelihood: -2.9491e+05
No. Observations: 61459 AIC: 5.898e+05
Df Residuals: 61452    BIC: 5.899e+05
Df Model: 6
Covariance Type: nonrobust
coef  std err  t    P>|t| [0.025 0.975]
const 0.4753  0.300  1.587  0.113 -0.112 1.063
x1    -0.0303  0.034  -0.889  0.374 -0.097 0.037
x2    -0.0429  0.664  -0.065  0.948 -1.344 1.258
x3     4.1931  0.736   5.697  0.000  2.751  5.636
x4    -0.0220  0.350  -0.063  0.950 -0.708 0.664
x5    -0.2675  0.616  -0.434  0.664 -1.474 0.939
x6    -0.2712  0.281  -0.967  0.334 -0.821 0.279
Omnibus: 287298.349 Durbin-Watson: 2.000
Prob(Omnibus): 0.000 Jarque-Bera (JB): 2269403007277.479
Skew: 171.156      Prob(JB): 0.00
Kurtosis: 29770.336 Cond. No. 48.0
```

Figure 13: OLS Regression Analysis

7.3 Predictive Modelling of 30-Day Readmissions

One reason logistic regression fits well here involves the outcome being simply yes or no - readmission within thirty days or not. Predicted probabilities emerge naturally from this method, allowing teams to sort cases by likelihood instead of applying one rule for everyone.

Starting with its foundation in clear statistical theory, the model handles both numbered inputs and grouped categories without trouble. When data dimensions stay small compared to patient counts, performance tends to remain stable. Unlike black-box algorithms, this technique shows how each factor shifts the odds, making choices easier to discuss with medical staff. Clarity matters especially when decisions affect treatment paths or resource planning. Because results link directly to familiar measures like odds ratios, trust in outputs grows among clinicians reviewing them. Simplicity, paired with consistent behavior across different samples, supports long-term use in real hospital systems. Ultimately, balancing accuracy and openness makes this approach hard to replace despite newer alternatives existing.

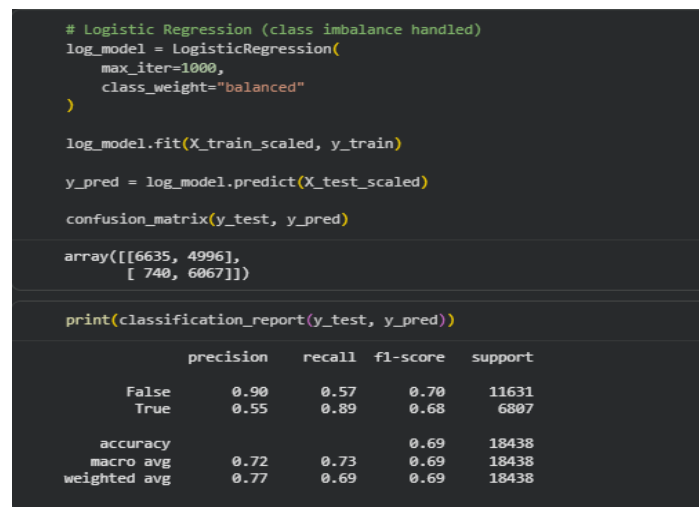


Figure 14: Logistic Regression Classification Report

Because the healthcare organisation focused on immediate practical needs, model assessment leaned into those goals. What mattered most was recall: how many actual readmissions the system caught. Overlooking someone truly at risk could lead to worsening health, unplanned ER visits, and more strain on hospital beds. When precision dropped, it simply meant extra check-ins for people who might not return - still manageable within resource limits. Putting recall first fits well with spotting at-risk patients early, which strengthens prevention efforts. Because the aim is catching more cases before they worsen, some incorrect alerts become acceptable trade-offs. That balance shifts emphasis toward keeping people safe while helping systems cope under pressure. The approach works best when avoiding missed cases matters most.

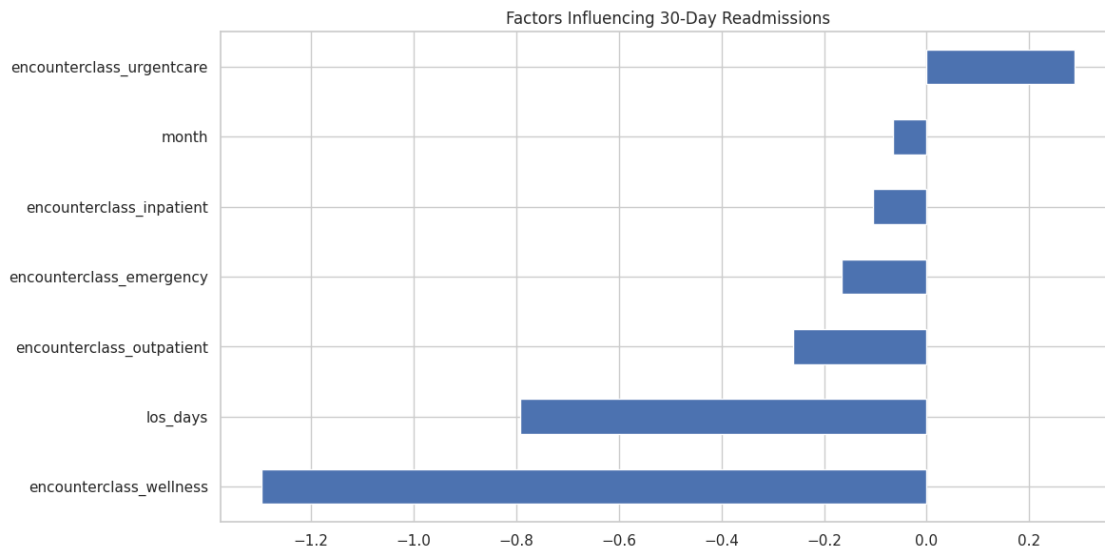


Figure 15: Feature Importance Analysis

8. Limitations and Ethical Considerations

Although this analysis is very insightful, there are several limitations that can be noted. The dataset is historical encounter-level data and might not adequately reflect all the clinical, social, or other behavioural determinants of length of stay and readmission risk. Other important variables like comorbidities, severity of the disease, socio-economic status, and post-discharge support were not accessible, which limited the power of explaining and predicting variables of the statistical and predictive model. Besides, there was a relative level of broadness in the categories of diagnosis, which might conceal heterogeneity within clinical groups. The predictive modelling part was an exploratory modelling that has not been externally validated and hence, its findings should be carefully mined and not applied in clinical decision-making without further substantiation.

Ethics are especially relevant to the analysis of healthcare data. Data analyses were anonymised, and no individual patient could be identified. The outcomes have been presented in aggregate mode to maintain privacy and confidentiality. It should also be ensured that predictive models do not accidentally support the current healthcare disparities or discriminate against the distribution of resources to vulnerable populations. It is thus necessary to have the communities of practice transparently report methods, limitations, and assumptions so as to ensure that there is responsible use of data-driven insights in the field of healthcare.

9. Conclusion

The report is an illustration of statistical programming as an effective tool to examine, describe, and forecast hospital utilisation patterns. In an integrative manner, using both exploratory analysis, inferential statistics, and predictive modelling, relevant drivers of bed-day utilisation,

seasonal pressure, and 30-day readmissions were found. The results focus on the significance of service-line variations, time, and length of stay in the process of determining operational demand.

These findings have direct practical implications for the hospital. The evidence of rising bed-day use and continued pressure on inpatient services suggests that the organisation will need to plan capacity more effectively, for example by adjusting staffing levels, allocating resources carefully, and preparing for higher demand during busy months. The patterns observed in patient journeys and early readmissions show that improvements in discharge planning are important, particularly to ensure that patients at higher risk receive proper information and support when they leave the hospital. In addition, the predictive model for readmissions offers a basis for more targeted follow-up, helping staff to focus monitoring and aftercare on the patients most likely to return. Taken together, these actions can help the hospital manage fluctuating demand, reduce avoidable readmissions, and maintain the quality of care.

References

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Code Documentation

Import Required Libraries

```
# importing libraries
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from datetime import timedelta
import matplotlib.dates as mdates
```

```
# Set visual style for professional plots
sns.set_theme(style="whitegrid")
plt.rcParams["figure.figsize"] = (12, 6)
```

Load and Inspect Dataset

```
# Load dataset
df = pd.read_csv("synthea_healthcaredata.csv")
```

```
# Initial inspection
df.head()
```

```
# Dataset structure and data types
df.info()
```

```
# Summary statistics for numerical columns
df.describe()
```

Data Cleaning & Preparation

```
# Standardise column names
df.columns = df.columns.str.strip().str.lower()

# Remove duplicated columns (keep first occurrence)
df = df.loc[:, ~df.columns.duplicated()]

# Verify column uniqueness
assert df.columns.is_unique

# Convert date columns safely
df["start"] = pd.to_datetime(df["start"], dayfirst=False, errors="coerce")
df["stop"] = pd.to_datetime(df["stop"], dayfirst=False, errors="coerce")

# Drop rows with invalid dates
df = df.dropna(subset=["start", "stop"])

# Compute Length of Stay (LOS) in days
df["los_days"] = (df["stop"] - df["start"]).dt.total_seconds() / (24 * 3600)

# Clip negative LOS (same-day or data issues)
df["los_days"] = df["los_days"].clip(lower=0)

# Create time features
df["year_month"] = df["start"].dt.to_period("M").astype(str)
df["month"] = df["start"].dt.month

# Final validation
df[["start", "stop", "los_days"]].describe()

# **Monthly Admissions, ED Visits, Bed-Days and Average LOS**

# Normalise encounter class
df["encounterclass"] = df["encounterclass"].astype(str).str.lower().str.strip()

# Flag ED encounters robustly
df["is_ed"] = df["encounterclass"].str.contains("emergency|ed", na=False)
```

```

# Monthly aggregation using PeriodIndex (chronologically safe)
monthly_metrics = (
    df.groupby(df["start"].dt.to_period("M"))
    .agg(
        admissions=("patient", "count"),
        ed_visits=("is_ed", "sum"),
        total_bed_days=("los_days", "sum"),
        avg_los=("los_days", "mean")
    )
    .reset_index()
)

monthly_metrics["year_month"] = monthly_metrics["start"].astype(str)
monthly_metrics.head()

# Plot admissions and ED visits

monthly_metrics["year_month"] = pd.to_datetime(monthly_metrics["year_month"])

fig, ax = plt.subplots(figsize=(12, 5))

ax.plot(monthly_metrics["year_month"],
        monthly_metrics["admissions"],
        label="Admissions")
ax.plot(monthly_metrics["year_month"],
        monthly_metrics["ed_visits"],
        label="ED Visits")

# major tick every 5 years instead of every year
ax.xaxis.set_major_locator(mdates.YearLocator(base=5))
ax.xaxis.set_major_formatter(mdates.DateFormatter('%Y'))

fig.autofmt_xdate(rotation=0) # keep labels horizontal

ax.set_title("Monthly Admissions and ED Visits")

```

```
ax.set_xlabel("Year")
ax.set_ylabel("Count")
ax.legend()
plt.tight_layout()
plt.show()
```

Plot bed-days and average LOS

```
fig, ax1 = plt.subplots()
```

```
ax1.plot(monthly_metrics["year_month"], monthly_metrics["total_bed_days"])
ax1.set_ylabel("Total Bed-Days")
```

```
ax2 = ax1.twinx()
ax2.plot(monthly_metrics["year_month"], monthly_metrics["avg_los"])
ax2.set_ylabel("Average LOS (days)")
```

```
plt.xticks(rotation=90)
plt.title("Monthly Bed-Days and Average LOS")
plt.tight_layout()
plt.show()
```

Service Lines & Diagnosis, Bed-Day Contribution Over Time

Service line bed-days over time

```
service_beddays["start"] = pd.to_datetime(service_beddays["start"])
service_beddays["year_month"] = service_beddays["start"]
```

```
fig, ax = plt.subplots(figsize=(12, 5))
```

```
sns.lineplot(
    data=service_beddays,
    x="year_month",
    y="total_bed_days",
```

```

    hue="encounterclass",
    ax=ax
)
ax.xaxis.set_major_locator(mdates.YearLocator(base=5))
ax.xaxis.set_major_formatter(mdates.DateFormatter('%Y'))

fig.autofmt_xdate(rotation=0)
ax.set_title("Bed-Days by Service Line Over Time")
ax.set_xlabel("Year")
ax.set_ylabel("Total Bed Days")
plt.tight_layout()
plt.show()

# Diagnosis categories with highest total bed-days
df["reasondescription"] = df["reasondescription"].fillna("Unknown")

top_diagnoses = (
    df.groupby("reasondescription")
      .agg(total_bed_days=("los_days", "sum"))
      .sort_values(by="total_bed_days", ascending=False)
      .head(10)
)

top_diagnoses

sns.barplot(
    x=top_diagnoses["total_bed_days"],
    y=top_diagnoses.index
)
plt.title("Top 10 Diagnosis Categories by Total Bed-Days")
plt.xlabel("Total Bed-Days")
plt.ylabel("Diagnosis")
plt.tight_layout()
plt.show()

# **Seasonal Variation by Month**

```

```
# Monthly seasonality for top diagnoses
seasonal_df = df[df["reasondescription"].isin(top_diagnoses.index)]

seasonal_utilisation = (
    seasonal_df.groupby(["month", "reasondescription"])
        .agg(total_bed_days=("los_days", "sum"))
        .reset_index()
)

sns.lineplot(
    data=seasonal_utilisation,
    x="month",
    y="total_bed_days",
    hue="reasondescription"
)

plt.title("Seasonal Utilisation Patterns by Diagnosis")
plt.xlabel("Month")
plt.ylabel("Total Bed-Days")
plt.tight_layout()
plt.show()

# **Patient Journeys & 30-Day Readmission Analysis**

# Sort encounters correctly
df = df.sort_values(["patient", "start"])

# Next encounter per patient
df["next_start"] = df.groupby("patient")["start"].shift(-1)

# Time to next encounter (days)
df["days_to_next"] = (
    df["next_start"] - df["stop"]
).dt.total_seconds() / (24 * 3600)

# 30-day readmission flag
df["readmitted_30_days"] = df["days_to_next"].between(0, 30, inclusive="both")
```

```
df["readmitted_30_days"].value_counts(normalize=True)

# Readmission rate by service line
readmission_summary = (
    df.groupby("encounterclass")
    .agg(
        encounters=("patient", "count"),
        readmissions=("readmitted_30_days", "sum")
    )
)

readmission_summary["readmission_rate"] = (
    readmission_summary["readmissions"] /
    readmission_summary["encounters"]
)

readmission_summary

sns.barplot(
    x=readmission_summary.index,
    y=readmission_summary["readmission_rate"]
)

plt.title("30-Day Readmission Rate by Service Line")
plt.ylabel("Readmission Rate")
plt.xlabel("Service Line")
plt.tight_layout()
plt.show()

# **Inferential Statistics (statsmodels)**

import statsmodels.api as sm
from scipy import stats
from statsmodels.tsa.seasonal import seasonal_decompose

# Seasonal decomposition of bed-days
ts_beddays = (
```

```

monthly_metrics
.set_index(pd.to_datetime(monthly_metrics["year_month"]))
["total_bed_days"]
)

decomposition = seasonal_decompose(ts_beddays, model="additive", period=12)
decomposition.plot()
plt.suptitle("Seasonal Decomposition of Total Bed-Days")
plt.tight_layout()
plt.show()

# ANOVA: Does LOS differ by month?
monthly_los_groups = [
    df[df["month"] == m]["los_days"] for m in range(1, 13)
]

stats.f_oneway(*monthly_los_groups)

# -----
# OLS Regression: Drivers of LOS (FIXED)
# -----

# Select variables
reg_df = df[["los_days", "encounterclass", "month"]].copy()

# Dummy encode categorical variable
reg_df = pd.get_dummies(reg_df, drop_first=True)

# Drop missing values
reg_df = reg_df.dropna()

# Separate dependent and independent variables
y = reg_df["los_days"].astype(float).to_numpy()
X = reg_df.drop("los_days", axis=1)

# Convert ALL predictors to float NumPy array
X = X.astype(float).to_numpy()

```



```
# Add constant manually (NumPy-safe)
X = sm.add_constant(X)

# Fit OLS model
ols_model = sm.OLS(y, X).fit()

# Show results
ols_model.summary()

# **Predictive Modelling (Scikit-Learn)**

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report, confusion_matrix

# Prepare ML dataset
ml_df = df[[
    "los_days",
    "encounterclass",
    "month",
    "readmitted_30_days"
]].dropna()

ml_df = pd.get_dummies(ml_df, drop_first=True)

X = ml_df.drop("readmitted_30_days", axis=1)
y = ml_df["readmitted_30_days"]

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.3,
    random_state=42,
    stratify=y
)
```

Scale features

```

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

```

Logistic Regression (class imbalance handled)

```

log_model = LogisticRegression(
    max_iter=1000,
    class_weight="balanced"
)

```

```
log_model.fit(X_train_scaled, y_train)
```

```
y_pred = log_model.predict(X_test_scaled)
```

```
confusion_matrix(y_test, y_pred)
```

```
print(classification_report(y_test, y_pred))
```

Feature importance

```

importance = pd.Series(
    log_model.coef_[0],
    index=X.columns
).sort_values()

```

```

importance.plot(kind="barh")
plt.title("Factors Influencing 30-Day Readmissions")
plt.tight_layout()
plt.show()

```